Program: **Diploma in Integrated Circuit Design and Fabrication**	
Course Code: **5379A**	Course Title: **Introduction to PERL Lab**
Semester: **5**	Credits: **1**
Course Category: Program Elective	
Periods per week: **4 (L:0 T:1 P:3)**	Periods per semester: **60**

Course Objectives:

- To provide fundamental concepts of PERL programming language.
- Provide hands-on programming experience through exercises while equipping students with the skills to perform file input/output operations in PERL.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Engineering Mathematics principles	1002 2002	Mathematics-I Mathematics-II	1 & 2
Basic Programming concepts	1008 3379	Introduction to IT systems lab Python Programming Lab	1&3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Develop PERL programs using standard I/O functions and basic operators	14	Applying
CO2	Develop PERL programs using control flow and looping statements	16	Applying
CO3	Construct PERL programs using subroutines.	15	Applying
CO4	Develop PERL programs using file I/O handling functions	12	Applying
	Lab Exam	3	Applying

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3		2	3			
CO3	3		2	3			
CO4	3		2	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Develop PERL programs using standard I/O functions and basic operators**		
M1.02	Install PERL Environment.	1P	Applying
M1.03	Demonstrate Input and Output Functions in PERL	1T+3P	Applying
M1.04	Demonstrating Standard Data Types (Scalars, Arrays, Hashes) in PERL	1T+3P	Applying
M1.05	Develop simple programs using basic operators. (Arithmetic operators, Comparison operators, Assignment operators, Logical operators, Bitwise operators.)	2T+3P	Applying
CO2	**Develop PERL programs using control flow and looping statements.**		
M2.01	Build programs using flow control or decision-making statements - IF statement, IF …ELSIF…ELSE and nested IF statements. **Suggested Experiments:** 1. Check whether the given two numbers are equal 2. Check whether the given number is odd or even 3. Find largest among the given three numbers 4. Character name of the day	3T+4P	Applying
M2.02	Build programs using loop statements -until, for, while, do while and nested loop. **Suggested Experiments:**	3T+6P	Applying

	1. Display the first n natural numbers 2. Read 10 numbers from keyboard and find their sum and average 3. Display the cube of the numbers upto the given integer 4. Display the multiplication table of a given integer. 5. Sum of the elements in a matrix.		
	Lab Exam 1	1.5	Applying
CO3	**Construct PERL programs using subroutines.**		
M3.01	Develop simple programs to demonstrate how to define and call subroutines.	3T+3P	Applying
M3.02	Develop programs using subroutines with and without arguments (including keyword-like arguments and parameters with default values) and with or without return values. **Suggested Experiments:** 1. Subroutine to Check Whether the Given Number is Even or Odd 2. Find the Sum and Average of Two Integer Numbers Using Subroutines (with Default Arguments) 3. Print Multiplication Table of a Given Number Using a Subroutine	3T+6P	Applying
CO4	**Develop PERL programs using file I/O handling functions**		
M4.01	Implement File I/O handling functions – open (), close (), read (), print (), getc ().	3T+6P	Applying
	Open Ended Experiments	3	
	Lab Exam 2	1.5	

** - Suggested Open Ended Projects

(Not for End Semester Examination but compulsory to be included in Continuous Internal

Evaluation. Students can-do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups. This experiment shall be included in the bona- fide record.

Example:

● Develop applications such as Make a simple calculator, feet to meters conversion

tool etc.

Text / Reference:

T/R	Book Title/Author
T1	Learning PERL, 6th Edition by Randal L. Schwartz, brian d foy, Tom Phoenix Released June 2011
T2	Tom Christiansen, Brian D Foy, Larry Wall, Jon Orwant, Programming PERL, O'Reily, 3rd Edition, 2010.
T3	PERL Programming for Beginners: An Introduction to Learn PERL Programming with Tutorials and Hands-On Examples Metzler, Nathan

Online resources:

Sl.No.	Website Link
1	https://www.tutorialspoint.com/PERL/index.htm
2	https://onlinecourses.swayam2.ac.in/aic20_sp31/preview
3	https://www.javatpoint.com/perl-tutorial